

Learning Holistic Whole-Body Loco-Manipulation with a Bipedal Mobile Manipulator

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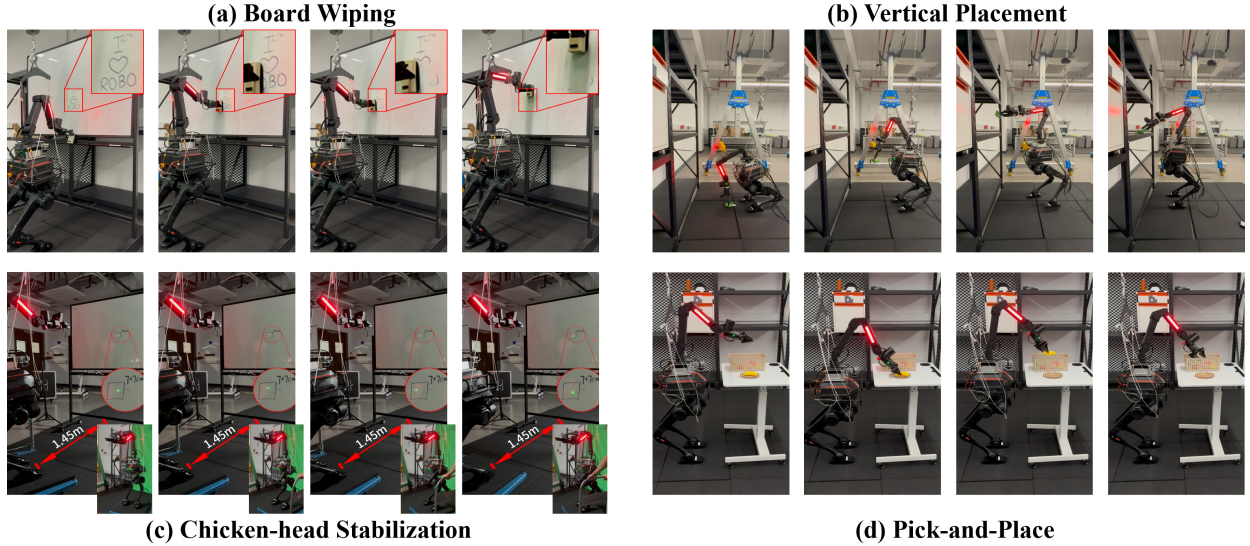


Fig. 1: Overview of our controller. Given only a 6-DoF end-effector target, the learned whole-body controller coordinates arm and leg motions to produce diverse whole-body manipulation behaviors.

Abstract—Bipedal loco-manipulation enables robots to interact with objects beyond the nominal workspace of their arms by coordinating locomotion and manipulation. Realizing this capability requires a low-level whole-body controller that translates task-level manipulation goals into coordinated arm and leg motions while maintaining balance. We present a unified whole-body controller trained with reinforcement learning that directly maps 6-DoF end-effector targets to coordinated actions for the bipedal base and robotic arm. Given only an end-effector target, the learned controller autonomously coordinates reaching, postural adaptation, and stepping without explicit base-velocity or footstep commands. A reward-gating strategy regulates the trade-offs among end-effector tracking, locomotion, and balance during training, while a temporal context estimator combines windowed Transformer encoding, recurrent GRU memory, and auxiliary dynamics prediction to extract dynamics-relevant information from observation history. Real-robot experiments demonstrate that the same controller supports reaching, postural adaptation, and stepping under commands from VR teleoperation, a learned diffusion policy, and scripted trajectories, providing a common end-effector interface for diverse manipulation tasks.

I. INTRODUCTION

Manipulation tasks often involve reaching objects at different heights and locations, including targets beyond the nominal workspace of a fixed-base arm. Loco-manipulation extends the robot’s reach by coordinating arm motion with body posture adjustment and locomotion [1], [2]. Effective coordination allows the robot to use its mobility in service of

manipulation, reaching targets that would otherwise remain inaccessible to the arm alone.

We adopt a bipedal mobile manipulator that combines a two-legged base with a single six-joint robotic arm. Its low hardware cost and fourteen controlled joints make it a minimalist platform for studying single-arm loco-manipulation without requiring a full humanoid upper body. Despite this structural simplicity, coordinating manipulation with bipedal mobility remains challenging: arm motion and postural adjustment must be coordinated with stepping while maintaining dynamic balance. This motivates a whole-body controller that uses the available arm and leg motions to accomplish manipulation goals.

For reuse across manipulation tasks, this controller should accept task-level goals and determine the required whole-body motion. The motion needed to reach an end-effector target depends on the robot’s current configuration: arm motion may suffice for one target, whereas another may require postural adjustment or stepping. Requiring human operators or high-level policies to specify these body motions ties manipulation commands to the details of whole-body coordination. An end-effector pose provides a geometrically interpretable command that specifies the manipulation objective while leaving whole-body motion generation to the controller. This separation allows different high-level command sources to share a common execution interface.

Learning-based whole-body control has enabled coordinated locomotion and manipulation on legged platforms [1], [3], [4]. UMI-on-Legs connects manipulation policies to a quadrupedal controller through end-effector trajectories [2]. Jiang et al. demonstrate direct 6-DoF end-effector pose tracking on a wheeled-quadrupedal manipulator using a nonlinear reward fusion module (RFM) [5]. For humanoids, ULC jointly tracks root, torso, and arm commands [6], while CEER provides an end-effector-root interface for heterogeneous high-level planners [7]. Building on these advances, we study whole-body control of a bipedal single-arm platform using a single instantaneous 6-DoF end-effector target as the sole task-level command. The controller must determine the required postural adjustment and stepping without additional locomotion or body-posture commands.

This formulation presents two coupled challenges. First, end-effector tracking, locomotion, and balance impose competing requirements on the same system. The controller must learn when to recruit the lower body to support reaching without sacrificing stability to reduce tracking error. Second, deployment under partial observations requires the controller to infer dynamics-relevant information from observation history. Coordinated whole-body responses therefore depend on both the regulation of training objectives and the representation of temporal information.

To address these challenges, we present a controller trained with reinforcement learning for end-effector-driven whole-body control of a bipedal manipulator. Building on RFM [5], we introduce phase-specific safety scores with lower-bounded gating and a best-so-far progress reward to coordinate locomotion, balance, and end-effector tracking. We further develop a temporal context estimator that combines windowed Transformer encoding, recurrent GRU memory, and auxiliary dynamics prediction to extract dynamics-relevant information from proprioceptive history. The resulting policy uses deployable observations and a single end-effector target to generate coordinated arm and leg actions, including reaching, postural adaptation, and stepping.

We integrate our controller into a real bipedal manipulation system with onboard policy execution, state estimation, and a common end-effector command representation. Commands from VR teleoperation, a learned diffusion policy, and scripted trajectories are transformed into this shared representation and executed by the same whole-body policy. Simulation comparisons and ablations evaluate task success, tracking accuracy, and action smoothness. Real-robot experiments demonstrate manipulation across these command sources and evaluate vertical end-effector reachability against a floating-base inverse-kinematics controller. These evaluations examine both the learning components and the controller’s use as a shared manipulation execution layer. Our main contributions are threefold:

- We formulate and realize end-effector-driven whole-body control for a bipedal manipulator, using a single 6-DoF target as the sole task-level command to coordinate arm motion, postural adaptation, and stepping without explicit locomotion or body-posture commands.

- We develop a unified learning framework for bipedal whole-body manipulation that combines phase-aware reward gating with a Transformer-GRU temporal context estimator. Simulation ablations demonstrate that both components contribute to end-effector tracking and whole-body stability.
- We integrate and deploy the controller on a low-cost bipedal manipulation system, demonstrating reuse across VR teleoperation, a learned diffusion policy, and scripted trajectories, together with expanded vertical reachability relative to a floating-base inverse-kinematics baseline.

II. RELATED WORK

A. Learning-Based Whole-Body Loco-Manipulation

Learning-based whole-body control coordinates locomotion and manipulation on legged platforms. Fu et al. [1] demonstrate unified arm-leg control on a quadrupedal manipulator, extending end-effector reach through arm-body coordination. Liu et al. [3] further investigate visual whole-body control for legged manipulation. Humanoid approaches organize coordination through unified, modular, or hierarchical control. ULC [6] learns a single whole-body tracking policy, whereas SEEC [8] learns upper-body compensation for locomotion-induced disturbances and supports different lower-body controllers. HiWET [9] separates high-level world-frame reasoning from low-level dynamic execution.

In contrast to prior work centered on quadrupedal manipulators or full humanoids, we study a complementary point in this design space: holistic loco-manipulation on a minimalist bipedal single-arm platform that expands task-relevant end-effector reachability through coordinated postural adaptation and stepping, where a single policy jointly controls all 14 arm and leg joints, rather than decomposing upper- and lower-body control, and resolves the resulting coordination problem under bipedal balance constraints.

B. Task-Space Interfaces for Loco-Manipulation

Task-space interfaces determine how motion decisions are divided between high-level command sources and low-level controllers. Portela et al. [4] learn 6-DoF end-effector pose tracking but handle locomotion with a separate policy. UMI-on-Legs [2] instead connects manipulation policies to a quadrupedal whole-body controller through task-frame end-effector trajectories. Other approaches retain explicit body-motion commands. ULC [6] tracks root, torso, and arm commands, while CEER [7] provides an end-effector-root interface for high-level planners.

Most closely related, Jiang et al. [5] directly track 6-DoF end-effector poses on a wheeled-quadrupedal manipulator without floating-base commands. Our reward-gating formulation builds on their RFM in the bipedal single-arm setting.

Against this landscape, our controller couples an end-effector-only task interface with holistic whole-body control. Given a single instantaneous 6-DoF end-effector target, the policy coordinates arm motion, postural adaptation, and stepping without explicit base or gait commands. The same

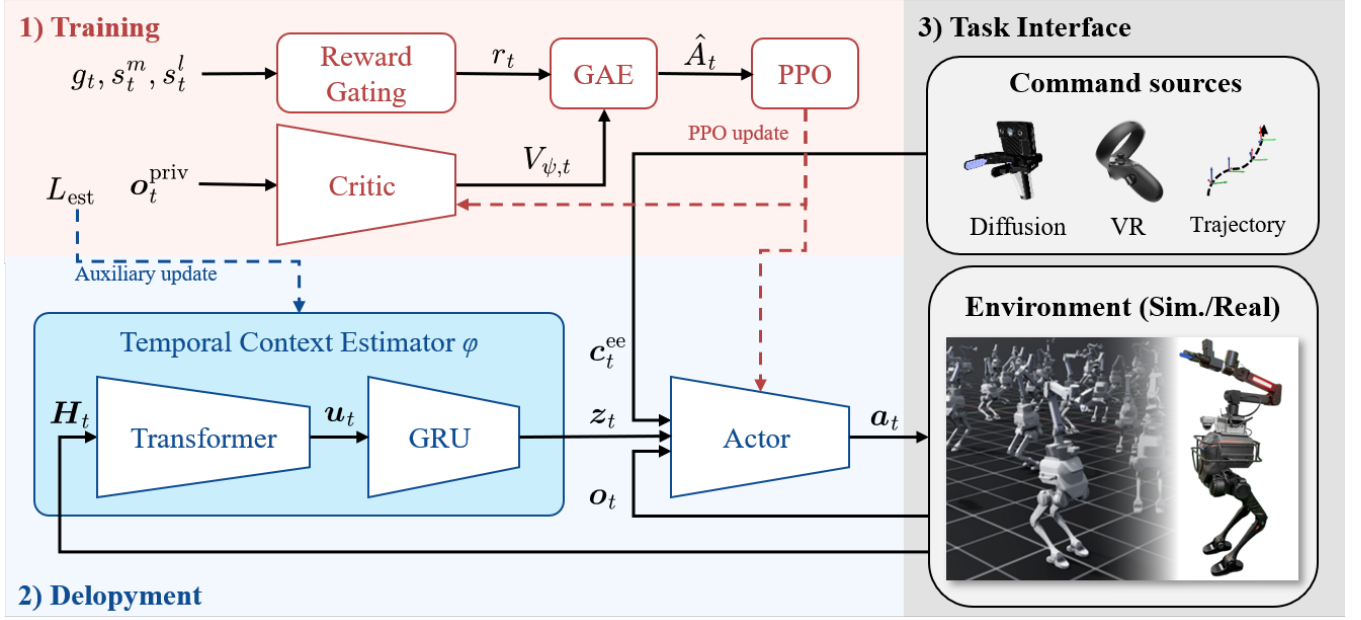


Fig. 2: Overview of the proposed controller. Given a 6-DoF end-effector target and proprioceptive observations, the policy coordinates whole-body actions through reward-gated training and temporal context estimation.

controller can therefore execute targets generated by VR teleoperation, learned manipulation policies, and scripted trajectories.

III. METHOD

Our goal is to learn a whole-body controller that enables a bipedal manipulator to track a desired end-effector pose while maintaining balance. Given the target pose and robot observations, the controller outputs joint-position commands for both the arm and legs. We combine temporal context estimation with reward-gated training to account for partially observed dynamics and balance the competing demands of tracking and whole-body motion.

A. Training Pipeline

Figure 2 illustrates the complete training pipeline. At each control step, the desired end-effector pose is expressed in the instantaneous base frame. A temporal estimator encodes recent observations using a Transformer and carries context across successive windows using a GRU. It produces a base linear-velocity estimate and a dynamics latent, which form the context $\mathbf{z}_t$. The actor combines this context with the current observation $\mathbf{o}_t$ and end-effector command $\mathbf{c}_t^{ee}$ to generate whole-body actions:

$$\begin{aligned} \mathbf{c}_t^{ee} &= [(\mathbf{p}_{BE,t}^*)^\top, \rho_6(\mathbf{R}_{BE,t}^*)^\top]^\top, \\ \mathbf{a}_t &\sim \pi_\theta(\cdot | \mathbf{o}_t, \mathbf{c}_t^{ee}, \mathbf{z}_t), \\ \mathbf{q}_t^{\text{des}} &= \mathbf{q}^{\text{default}} + \mathbf{s} \odot \mathbf{a}_t. \end{aligned} \quad (1)$$

Here, $\mathbf{p}_{BE,t}^*$ and $\mathbf{R}_{BE,t}^*$ are the desired end-effector position and orientation in the base frame. The mapping ρ_6 concatenates the first two columns of the rotation matrix to obtain a continuous 6D representation [10], giving $\mathbf{c}_t^{ee} \in \mathbb{R}^9$. Table I

lists the 58-dimensional observation and the context inputs. The task-progress reference d_t^{ref} is generated internally. The end-effector pose is the sole external task command; no reference base velocity, gait phase, contact schedule, footstep command, or discrete behavior mode is provided.

The normalized action $\mathbf{a}_t \in \mathbb{R}^{14}$ contains position offsets for eight lower-body joints and six arm joints. Here, $\odot$ denotes element-wise multiplication and $\mathbf{s} \in \mathbb{R}^{14}$ contains joint-specific action scales. These scales are 0.6 for the lower body, 0.3 for arm joints J1–J3, and 0.2 for J4–J6. Low-level joint-position controllers track the resulting targets.

A scheduled SE(3)-distance reference defines a continuous locomotion–manipulation coefficient. The reward combines this coefficient with phase-specific safety scores to coordinate target approach, precise tracking, and whole-body stability. Lower-bounded gating preserves the learning signal during recovery, while a best-so-far progress term rewards further reductions in tracking error. Section III-C gives the complete formulation.

During training, we sample 6-DoF end-effector targets throughout the robot’s whole-body workspace, including targets reachable through arm motion alone and targets that require postural adaptation or stepping. We apply domain randomization to the friction coefficients, base mass, initial base state, and initial leg-joint configuration. We additionally apply periodic velocity perturbations and uniform noise to the proprioceptive and end-effector observations.

The resulting rollouts are used to update an asymmetric actor–critic with PPO [11], [12]. The actor uses only the deployable inputs listed in Table I. The critic additionally receives base linear velocity, joint torques and accelerations, base height, end-effector velocities, foot positions, control gains, base mass, and contact parameters for value estima-

TABLE I: Deployable inputs to the controller.

Input term	Symbol	Dimension
<i>Robot observation $\mathbf{o}_t$</i>		
Base angular velocity	$\boldsymbol{\omega}_t^B$	3
Projected gravity	$\mathbf{g}_t^B$	3
Joint positions	$\mathbf{q}_t^{\text{obs}}$	14
Joint velocities	$\dot{\mathbf{q}}_t$	14
Previous action	$\mathbf{a}_{t-1}$	14
End-effector pose	$\mathbf{x}_t^{\text{ee}}$	9
Task-progress reference	d_t^{ref}	1
<i>Task command</i>		
End-effector target	$\mathbf{c}_t^{\text{ee}}$	9
<i>Temporal context</i>		
Estimated base velocity	$\hat{\mathbf{v}}_t$	3
Dynamics latent	$\boldsymbol{\xi}_t$	64

tion. The temporal estimator is optimized through the auxiliary objectives defined in Sec. III-B. During deployment, only the actor and temporal estimator are retained.

B. Temporal Context Estimation

Variations in robot and environment dynamics alter the motion response produced by a given control action. Although these variations are not directly observable, their effects appear in the temporal evolution of the robot state. We therefore infer a dynamics-relevant context from recent onboard observations and use it to condition the whole-body control policy.

At control step t , the estimator receives the most recent $H = 10$ observations:

$$\mathcal{H}_t = [\mathbf{h}_{t-H+1}, \dots, \mathbf{h}_t]^\top \in \mathbb{R}^{H \times 57}. \quad (2)$$

Each $\mathbf{h}_t \in \mathbb{R}^{57}$ contains base angular velocity, projected gravity, joint positions and velocities, estimated joint torques, and the measured end-effector pose. All components are available from onboard sensing or state estimation at deployment. We encode this history with a Transformer [13]:

$$\mathbf{u}_t = f_{\text{TF}, \phi_{\text{TF}}}(\mathcal{H}_t). \quad (3)$$

Positional encodings preserve temporal order, while self-attention models relationships among robot responses at different steps within the window. A GRU [14] then propagates information across consecutive windows:

$$(\mathbf{r}_t, \mathbf{m}_t) = f_{\text{GRU}, \phi_{\text{GRU}}}(\mathbf{u}_t, \mathbf{m}_{t-1}), \quad (4)$$

where $\mathbf{m}_t$ is the recurrent state and $\mathbf{r}_t$ is the corresponding GRU output.

The estimator combines the GRU output with the Transformer representation through the residual connection implemented in the network:

$$\mathbf{e}_t = \mathbf{u}_t + f_{\text{out}}(\mathbf{r}_t). \quad (5)$$

We partition the resulting estimator output as

$$\mathbf{e}_t = [\hat{\mathbf{v}}_t^\top, \boldsymbol{\mu}_t^\top, (\log \boldsymbol{\sigma}_t^2)^\top]^\top \in \mathbb{R}^{131}, \quad (6)$$

where $\hat{\mathbf{v}}_t \in \mathbb{R}^3$ estimates the local base linear velocity, and $\boldsymbol{\mu}_t, \log \boldsymbol{\sigma}_t^2 \in \mathbb{R}^{64}$ parameterize a diagonal Gaussian distribution. Using the reparameterization trick [15], we then sample the latent and construct the policy context:

$$\begin{aligned} \boldsymbol{\sigma}_t &= \sqrt{\exp(\log \boldsymbol{\sigma}_t^2) + 10^{-4}}, \\ \boldsymbol{\xi}_t &= \boldsymbol{\mu}_t + \boldsymbol{\sigma}_t \odot \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \\ \mathbf{z}_t &= [\hat{\mathbf{v}}_t^\top, \boldsymbol{\xi}_t^\top]^\top \in \mathbb{R}^{67}. \end{aligned} \quad (7)$$

The context $\mathbf{z}_t$ is supplied to the actor together with the current observation and end-effector command, as defined in Sec. III-A.

During training, an MLP decoder predicts the next-step target observation from the sampled latent:

$$\hat{\mathbf{y}}_{t+1} = f_{\text{dec}}(\boldsymbol{\xi}_t). \quad (8)$$

The target $\mathbf{y}_{t+1} \in \mathbb{R}^{49}$ is the configured next-observation vector. It contains base angular velocity, projected gravity, joint positions and velocities, end-effector pose, and foot positions. It is distinct from the actor observation $\mathbf{o}_{t+1}$. Together with velocity supervision and latent regularization, this prediction defines the estimator objective:

$$\begin{aligned} \mathcal{L}_{\text{est}} &= \|\hat{\mathbf{v}}_t - \mathbf{v}_t\|_1 + \|\hat{\mathbf{y}}_{t+1} - \mathbf{y}_{t+1}\|_1 \\ &\quad + \beta D_{\text{KL}}[\mathcal{N}(\boldsymbol{\mu}_t, \text{diag}(\boldsymbol{\sigma}_t^2)) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})]. \end{aligned} \quad (9)$$

We set the KL weight to $\beta = 0.1$. The context is detached during PPO updates, so the estimator receives gradients only from these auxiliary objectives. The prediction objectives encourage the estimator to encode the robot's recent motion response, which provides the actor with information relevant to dynamics adaptation. The recurrent state is reset at the beginning of each episode.

C. Biped-Aware Reward Gating

Let e_t^p and e_t^R denote the end-effector position and geodesic orientation errors. The reference and phase variable follow RFM [5]:

$$\begin{aligned} d_0^{\text{ref}} &= 2e_0^p + e_0^R, \\ d_t^{\text{ref}} &= \max(d_{t-1}^{\text{ref}} - v^{\text{ref}} \Delta t, 0), \\ g_t &= \sigma(k(d_t^{\text{ref}} - \mu)). \end{aligned} \quad (10)$$

Here, σ is the sigmoid function, v^{ref} sets the reference decay rate, and k and μ control the transition. The phase variable g_t is computed from a scheduled SE(3)-distance reference that decreases over the command duration. A large g_t emphasizes locomotion and whole-body target-approach rewards, whereas a small g_t emphasizes precise end-effector tracking and whole-body stabilization.

Locomotion and stationary manipulation admit different safe body motions. The locomotion score considers foot geometry, base height, and base orientation, allowing the motion needed for stepping. The manipulation score additionally considers base, end-effector, and joint velocities, as

well as arm configuration, to suppress residual motion near the target. For each phase $q \in \{m, l\}$, we define

$$\begin{aligned} E_t^q &= \frac{1}{Z_q} \sum_{j \in C_q} \alpha_j^q \frac{\delta_{j,t}}{\tau_j}, \\ s_t^q &= \exp(-E_t^q / \sigma_s^2), \\ \tilde{s}_t^q &= s_t^q + c_s, \quad c_s = 0.4. \end{aligned} \quad (11)$$

Here, C_q denotes the corresponding set of safety quantities, with m denoting manipulation and l locomotion. Each deviation $\delta_{j,t}$ is normalized by tolerance τ_j , weighted by α_j^q , and aggregated using normalization constant Z_q . The parameter σ_s controls the decay of the safety score. A severe transient disturbance can drive s_t^q close to zero. Directly using this score as a multiplicative gate may suppress the task-learning signal needed for recovery. The offset c_s relaxes the gate to retain this signal. The relaxed score $\tilde{s}_t^q$ gates task rewards, while s_t^q enters the explicit safety reward.

RFM uses a cumulative-error term to regulate task progress. For bipedal target approach, we instead reward reductions relative to the best position and orientation errors already achieved for the current command. This formulation provides a positive signal only when the controller makes further tracking progress. Defining $\bar{e}_t^p = \min_{0 \leq j \leq t} e_j^p$ and $\bar{e}_t^R = \min_{0 \leq j \leq t} e_j^R$, we use

$$\begin{aligned} r_t^{\text{prog}} &= 2[\bar{e}_{t-1}^p - e_t^p]_+ \exp(-\bar{e}_t^p / \sigma_{\text{prog}}) \\ &\quad + [\bar{e}_{t-1}^R - e_t^R]_+ \exp(-\bar{e}_t^R / \sigma_{\text{prog}}), \end{aligned} \quad (12)$$

where $[x]_+ = \max(x, 0)$ and σ_{prog} sets the error scale. We retain RFM's position-orientation prioritization and reference-tracking design [5]. The position reward uses micro-enhancement to increase sensitivity near the target. A coarse orientation term acts throughout the command, while a fine orientation term is activated by small position error and the manipulation gate. Defining $e_t^{\text{ref}} = [|d_t^{\text{ref}} - d_t| - \gamma]_+$, the individual terms and their scales are summarized in Table II. The resulting task reward is

$$\begin{aligned} r_t^{\text{task}} &= w_p(1 - g_t)\tilde{s}_t^m r_t^p + w_{R,c} r_t^{R,c} \\ &\quad + w_{R,f}(1 - g_t)\tilde{s}_t^m r_t^{R,f} \\ &\quad + w_{\text{ref}} g_t \tilde{s}_t^l r_t^{\text{ref}} \\ &\quad + w_{\text{prog}} \tilde{s}_t^l (0.25 + 0.75g_t) r_t^{\text{prog}}. \end{aligned} \quad (13)$$

The complete reward adds the explicit safety reward and regularization terms:

$$\begin{aligned} r_t &= r_t^{\text{task}} + w_s [(1 - g_t)s_t^m + g_t s_t^l] \\ &\quad + \sum_i w_i r_{i,t}^{\text{reg}}. \end{aligned} \quad (14)$$

Here, the w terms are reward weights. The regularization terms penalize excessive torque, power, action variation, foot slip, non-flat support, insufficient leg separation, joint-limit violations, and unsafe termination.

TABLE II: Reward terms and scales used for policy training.

Term	Definition	Scale
<i>Tracking</i>		
Position tracking	$r_t^p = e^{-e_t^p / \sigma_p^2} + e^{-5e_t^p / \sigma_p^2}$	6.0
Coarse orientation	$r_t^{R,c} = \exp[-(e_t^R / 1.5)^2]$	6.0
Fine orientation	$r_t^{R,f} = e^{-e_t^p / 0.5} \exp[-(e_t^R / 0.25)^2]$	6.0
<i>Phase</i>		
Reference tracking	$r_t^{\text{ref}} = \exp(-e_t^{\text{ref}} / \sigma_{\text{ref}}^2)$	10.0
<i>Progress</i>		
Best-so-far progress	r_t^{prog} in Eq. (12)	15.0
<i>Safety</i>		
Phase safety	$(1 - g_t)s_t^m + g_t s_t^l$ in Eq. (14)	2.0
<i>Regularization</i>		
Joint torque	$\ \tau_t\ _2^2$	-1.0×10^{-5}
Joint power	$\sum_j \tau_{t,j} \dot{q}_{t,j} $	-2.0×10^{-5}
Action variation	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-2.0×10^{-2}
Foot slip	$\sum_{f \in \mathcal{F}_c} \ \mathbf{v}_{f,xy}\ _2^2$	-0.5
Non-flat support	$\sum_{f \in \mathcal{F}_c} \ \mathbf{z}_{f,xy}^w\ _2^2$	-1.0
Leg separation	$[d_{\min} - y_t^L - y_t^R]_+$	-10.0
Joint limits	$\sum_j ([q_{t,j} - q_j^{\max}]_+ + [q_j^{\min} - q_{t,j}]_+)$	-1.0
Unsafe termination	$\mathbf{1}_{\text{terminated}}$	-100

IV. EXPERIMENTS AND RESULTS

We evaluate our controller in both simulation and real-world deployment. In simulation, we conduct ablation studies to investigate the effects of reward-gated training and temporal context estimation on end-effector tracking and whole-body stability. We then deploy the learned policy on the real bipedal manipulator and demonstrate its ability to execute diverse manipulation tasks under different high-level command sources, including VR teleoperation, learned diffusion-policy commands, and scripted end-effector trajectories.

A. Experimental Setup

Our real-world platform, shown in Fig. 3, consists of a LimX TRON 1 biped equipped with an ARX L5 robotic arm, a GENROBOT UMI gripper [16], a Livox Mid-360 LiDAR, and an NVIDIA Jetson Orin NX. The Jetson runs onboard state estimation and the learned whole-body policy, which jointly controls the biped and robotic arm. FAST-LIO2 [17] provides real-time LiDAR-inertial localization, while LiDAR measurements also provide ground-height estimation. The estimated base pose and arm joint states are combined through forward kinematics to recover the end-effector pose in the world frame for tracking the commanded 6-DoF end-effector target.

We train our controller in Isaac Lab [18] using PPO [12] with 8,192 parallel environments for 20,000 policy iterations. Training on a single NVIDIA GeForce RTX 5090 GPU takes approximately 28 hours. The policy runs at 50 Hz, while the resulting joint-position commands are tracked at 500 Hz by the low-level PD controller.

B. Evaluation in Simulation

We evaluate the policy trained with our method in simulation to quantify task-level tracking performance, control smoothness, and the contribution of the two key design choices: reward gating and temporal context estimation. All

TABLE III: Simulation evaluation and ablation results.

Study	Method	Success $\uparrow$ (%)	Pos. Mean $\downarrow$ (cm)	Pos. P95 $\downarrow$ (cm)	Ori. Mean $\downarrow$ (deg)	Ori. P95 $\downarrow$ (deg)	Action Variation $\downarrow$
Reward	Matched additive	82.73 $\pm$ 0.93	3.23 $\pm$ 0.02	14.35 $\pm$ 0.25	2.12 $\pm$ 0.12	5.08 $\pm$ 0.59	0.218 $\pm$ 0.010
	Controller (reward gating)	88.30 $\pm$ 2.50	2.85 $\pm$ 0.10	5.38 $\pm$ 0.35	3.19 $\pm$ 0.04	6.49 $\pm$ 0.25	0.165 $\pm$ 0.006
Temporal context	No latent	69.87 $\pm$ 1.19	3.65 $\pm$ 0.07	7.51 $\pm$ 0.07	3.85 $\pm$ 0.08	7.58 $\pm$ 0.46	0.236 $\pm$ 0.004
	DreamWaQ/CENet	68.90 $\pm$ 2.16	3.83 $\pm$ 0.14	8.14 $\pm$ 0.58	4.22 $\pm$ 0.17	9.06 $\pm$ 0.76	0.169 $\pm$ 0.007
	GRU only	73.23 $\pm$ 3.72	3.59 $\pm$ 0.13	6.90 $\pm$ 0.28	3.42 $\pm$ 0.12	7.66 $\pm$ 0.40	0.173 $\pm$ 0.008
	Transformer only	84.43 $\pm$ 1.66	3.21 $\pm$ 0.14	6.33 $\pm$ 0.36	2.77 $\pm$ 0.04	5.23 $\pm$ 0.13	0.160 $\pm$ 0.006
	Controller (Transformer+GRU)	88.30 $\pm$ 2.50	2.85 $\pm$ 0.10	5.38 $\pm$ 0.35	3.19 $\pm$ 0.04	6.49 $\pm$ 0.25	0.165 $\pm$ 0.006
Upper bound	Privileged oracle	94.53 $\pm$ 1.54	2.51 $\pm$ 0.10	4.78 $\pm$ 0.24	3.14 $\pm$ 0.12	5.91 $\pm$ 0.36	0.112 $\pm$ 0.009

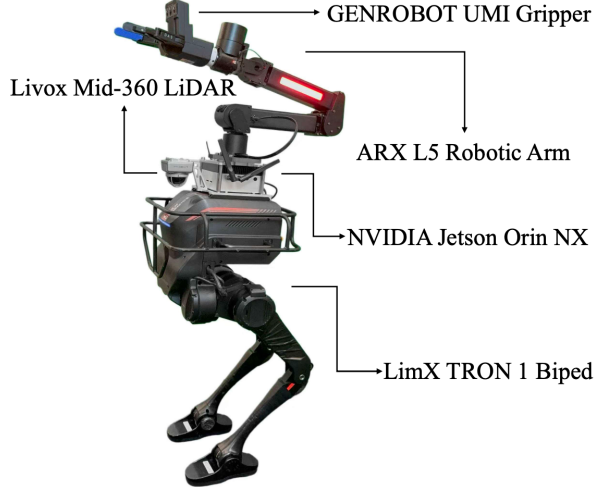


Fig. 3: Real-world platform consisting of a LimX TRON 1 biped, an ARX L5 robotic arm, a GENROBOT UMI gripper, a Livox Mid-360 LiDAR, and an NVIDIA Jetson Orin NX for onboard state estimation and whole-body policy inference.

methods are evaluated under the same target distribution, domain-randomization settings, episode duration, and success criteria.

Evaluation protocol and metrics. For each method, we train three policies using independent random seeds and evaluate the final checkpoint after 20,000 training iterations. Each policy is evaluated over 1,000 episodes under the same target distribution and domain-randomization settings, resulting in 3,000 evaluation episodes for each method. The results are reported as mean $\pm$ standard deviation over the three training seeds. We report task success rate together with the mean and 95th-percentile (P95) end-effector tracking errors. An episode is considered successful if the position and orientation errors remain below 5 cm and 7° , respectively, during the final 1 s of the rollout. Position error is measured as the Euclidean distance, while orientation error is computed as the geodesic distance between rotation matrices. At each step, action variation is computed as $\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_2^2$ and averaged over the evaluation rollouts; lower values indicate smoother control.

Reward-gating ablation. We first compare the proposed

reward-gating formulation with a matched additive reward containing the same reward terms and coefficients. As shown in Table III, reward gating increases the success rate from 82.73% to 88.30% and reduces the mean position error from 3.23 cm to 2.85 cm. The improvement is particularly pronounced in the tail of the error distribution: position P95 decreases from 14.35 cm to 5.38 cm. At the same time, action variation is reduced from 0.218 to 0.165, indicating smoother policy outputs.

Interestingly, the matched additive reward achieves lower orientation error than reward gating. This result highlights a trade-off between minimizing individual tracking terms and maximizing task-level whole-body performance. The additive objective can achieve accurate orientation tracking on successful rollouts, yet exhibits lower overall success and substantially larger tail position errors. In contrast, reward gating prioritizes coordinated whole-body adaptation when multiple objectives compete, resulting in higher task completion and fewer large tracking failures.

Temporal-context ablation. We next evaluate the contribution of temporal context using no-latent, CENet [19], GRU-only, Transformer-only, and the proposed Transformer-GRU estimator. Removing temporal context reduces the success rate to 69.87% and increases the action variation to 0.236. Introducing either recurrent or attention-based temporal encoding improves performance, showing that historical observations are important for inferring the latent state of the whole-body system.

The full Transformer-GRU estimator achieves the highest non-oracle success rate of 88.30%, together with the lowest mean and P95 position errors of 2.85 cm and 5.38 cm, respectively. Compared with the Transformer-only variant, the full estimator improves success by 3.87 percentage points and substantially reduces both mean and tail position errors, although the Transformer-only model attains slightly lower orientation error and action variation. These results suggest that attention over a finite observation window and recurrent memory provide complementary temporal information: the Transformer captures structured short-term dependencies, while the GRU maintains information beyond the fixed observation window.

Finally, the privileged oracle provides ground-truth simulation information to the actor and reaches a 94.53% success rate, serving as an approximate upper bound. The relatively

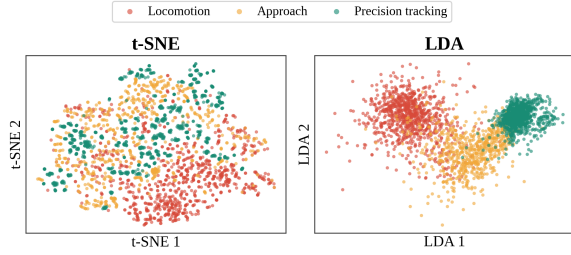


Fig. 4: Visualization of the learned temporal latent representation. Left: an unsupervised t-SNE projection of the latent features. Right: an LDA projection using task phases defined according to end-effector position error.

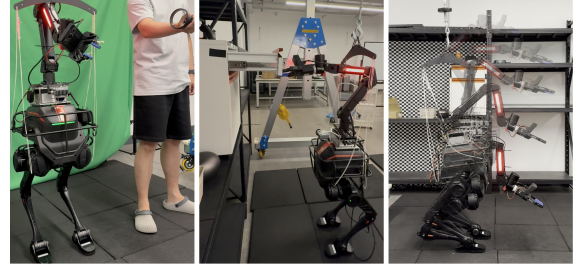


Fig. 5: Unified controller interface for real-world deployment. From left to right: Quest teleoperation, UMI-based diffusion policy, and scripted trajectories tracking.

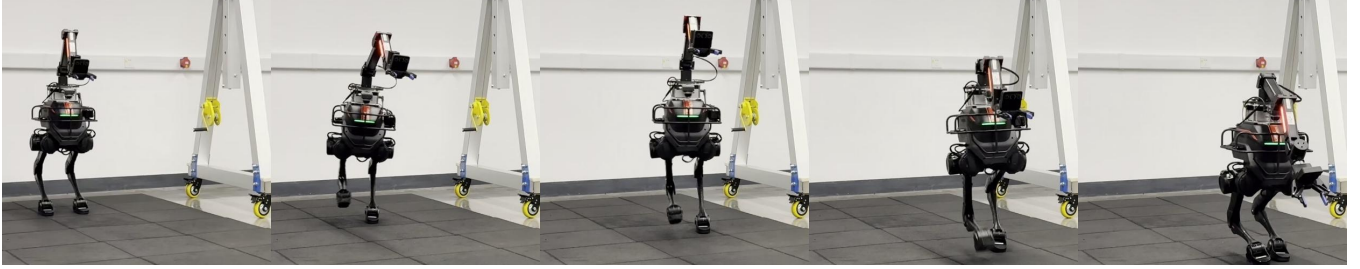


Fig. 6: Successive frames of real-world whole-body adaptation during end-effector tracking. The controller autonomously coordinates postural adjustment and stepping using only the commanded 6-DoF end-effector pose.

small gap in task success between our controller and the oracle suggests that the proposed temporal estimator recovers a substantial portion of the information required for whole-body adaptation from observation history alone.

Latent representation analysis. To examine the information encoded by the temporal context estimator, we visualize its latent features using t-SNE and linear discriminant analysis (LDA), as shown in Fig. 4. The samples are categorized into locomotion, approach, and precision tracking phases according to the current end-effector position error. The unsupervised t-SNE projection reveals locally structured and partially overlapping regions, whereas the supervised LDA projection more clearly exposes the task-phase-dependent organization of the representation. Importantly, the overlap between adjacent phases suggests a continuous transition rather than three completely independent motion classes.

C. Real-World Demonstrations

We evaluate our method on the real robot from two complementary perspectives: (i) whether it can provide a unified task-space interface for heterogeneous high-level command sources, and (ii) whether the learned whole-body behavior provides a larger and more stable manipulation workspace than a conventional floating-base + inverse-kinematics (IK) controller.

Unified task-space interface. As illustrated in Fig. 5, we drive the same controller using three distinct command sources: Quest-based human teleoperation, a diffusion policy trained from UMI-style demonstrations [16], [20], and scripted trajectories. Despite their different origins, commands from all high-level sources are transformed into

the instantaneous base frame before being passed to our controller, ensuring a common task-space representation independent of the command source. No source-specific base-velocity, gait, footstep, or whole-body joint commands are required, and the controller is used without modification across all three interfaces.

Across these interfaces, we demonstrate a diverse set of real-world manipulation tasks. With Quest-based teleoperation, the robot picks up a plush toy from the ground, takes a step, and places it onto a shelf, requiring coordinated squatting, reaching, locomotion, and recovery. We also teleoperate the robot to pick up a whiteboard eraser and wipe a blackboard, demonstrating continuous task-space motion while maintaining whole-body balance. With the UMI-based diffusion policy, the robot autonomously pushes a cabinet door closed using only the predicted end-effector commands. Finally, scripted trajectories command the end effector to follow repeated upward and downward motions, allowing us to evaluate whole-body adaptation to targets spanning different heights. Despite the different command sources and task requirements, all behaviors are executed by the same controller through the same 6-DoF end-effector interface. Figure 6 shows successive frames of the controller autonomously recruiting lower-body motion and stepping while tracking an end-effector target. Additional real-world demonstrations are provided in the supplementary material.

Whole-body workspace and stability. We further compare our controller with a floating-base + IK baseline to evaluate the benefit of learned whole-body coordination for task-space reachability. As shown in Fig. 7, the floating-base + IK controller achieves a vertical end-effector workspace

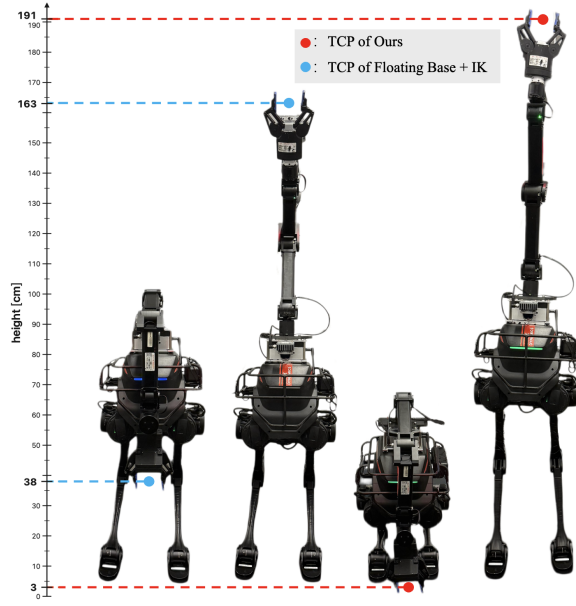


Fig. 7: Real-world comparison of vertical end-effector reachability between our controller and a floating-base + IK baseline. The controller exploits autonomous whole-body motion to extend the reachable height from 3 to 191 cm, compared with 38 to 163 cm for the baseline.

of approximately 38–163 cm. In comparison, our controller extends the reachable range to approximately 3–191 cm by autonomously coordinating squatting, standing extension, and arm motion. This substantially larger workspace is achieved without explicitly commanding the base posture or locomotion. Beyond reachability, we observe that the floating-base + IK baseline exhibits pronounced oscillatory motion near challenging workspace boundaries, whereas our controller maintains substantially smoother and more stable whole-body behavior. This comparison highlights that our controller does not merely increase geometric reachability, but learns to exploit whole-body mobility while preserving balance and coordinated end-effector motion.

V. CONCLUSION

We presented an end-effector-conditioned whole-body controller trained with reinforcement learning for bipedal mobile manipulation that directly maps a single 6-DoF end-effector target to coordinated whole-body actions without explicit base-velocity, gait, or footstep commands. The controller combines reward-gated training with a Transformer-GRU temporal context estimator to support accurate manipulation while maintaining dynamic balance. Experiments in simulation and on a real bipedal manipulator demonstrate that the same controller can autonomously produce arm-dominant reaching, postural adaptation, squatting, and stepping, while also serving as a common task-space interface for human, learned, and scripted high-level commands. Future work will extend the framework toward more dynamic loco-manipulation behaviors and improve task-space tracking

precision for fine-grained manipulation, particularly when following learned high-level commands.

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